Lab Exercise 16- Creating a ConfigMap from a Folder

In this lab exercise, you will learn how to create a ConfigMap in Kubernetes from a folder containing multiple files. This allows you to manage multiple configuration files as a single ConfigMap.

Step 1: Set Up Kubernetes Cluster

Ensure you have access to a Kubernetes cluster. You can use a local setup with Minikube, kind, or use a cloud-based Kubernetes service.

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl version --client
Client Version: v1.34.1
Kustomize Version: v5.7.1
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8>
```

Step 2: Prepare the Folder with Multiple Files

Create a directory named configmap-folder:

```
mkdir configmap-folder
cd configmap-folder/
```

Create multiple configuration files inside the directory:

config1.txt with content:

```
key1=value1
```

config2.txt with content:

```
key2=value2
```

config3.txt with content:

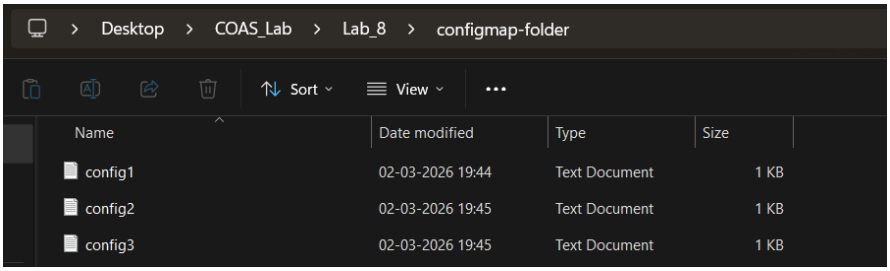
```
key3=value3
```

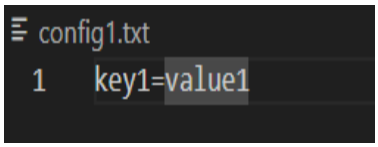
Verify the directory structure:

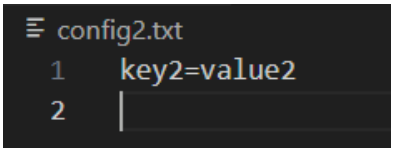
```
ls configmap-folder/
cd ..
```

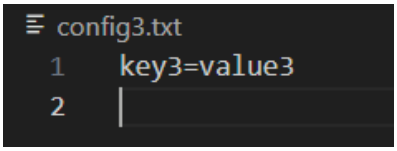
You should see the files config1.txt, config2.txt, and config3.txt.

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> mkdir configmap-folder

    Directory: C:\Users\HP\Desktop\COAS_Lab\Lab_8

Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d----          02-03-2026    19:43                configmap-folder

PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> cd configmap-folder/
```










```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8\configmap-folder> ls

    Directory: C:\Users\HP\Desktop\COAS_Lab\Lab_8\configmap-folder

Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a---          02-03-2026    19:44             13 config1.txt
-a---          02-03-2026    19:45             13 config2.txt
-a---          02-03-2026    19:45             13 config3.txt

PS C:\Users\HP\Desktop\COAS_Lab\Lab_8\configmap-folder>
```

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8\configmap-folder> cd ..
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8>
```

Step 3: Create a ConfigMap from the Folder

Create a ConfigMap named folder-configmap from the folder:

```
kubectl create configmap folder-configmap --from-file=configmap-folder/
```

Verify the ConfigMap:

```
kubectl get configmap folder-configmap -o yaml
```

You should see the folder-configmap with the contents of all the files in the configmap-folder.

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl create configmap folder-configmap --from-file=configmap-folder/
configmap/folder-configmap created
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl get configmap folder-configmap -o yaml
apiVersion: v1
data:
  config1.txt: "key1=value1\r\n"
  config2.txt: "key2=value2\r\n"
  config3.txt: "key3=value3\r\n"
kind: ConfigMap
metadata:
  creationTimestamp: "2026-03-02T14:19:15Z"
  name: folder-configmap
  namespace: default
  resourceVersion: "47862"
  uid: 6c76beed-8406-4685-a691-2ec571ed98f9
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8>
```

Step 4: Use the ConfigMap in a Pod

Create a file named **pod-folder-configmap.yaml** with the following content:

```
apiVersion: v1
kind: Pod
metadata:
  name: configmap-pod
spec:
  containers:
  - name: nginx
    image: nginx
    envFrom:
    - configMapRef:
        name: folder-configmap
```

In this manifest:

- The Pod mounts the folder-configmap at /etc/config.
- The busybox container concatenates and prints the contents of the files.

Apply the manifest to create the Pod:

```
kubectl apply -f pod-folder-configmap.yaml
```

Check the status of the Pod:

```
kubectl get pods
```

View the Pod logs to see the content of the mounted files:

```
kubectl exec -it configmap-pod -- printenv
```

You should see the contents of config1.txt, config2.txt, and config3.txt printed.

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl apply -f pod-folder-configmap.yaml
error: the path "pod-folder-configmap.yaml" does not exist
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl apply -f pod-folder-configmap.yaml
pod/configmap-pod created
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl get pods
NAME            READY   STATUS    RESTARTS   AGE
configmap-pod   1/1     Running   0          10s
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl get pods
NAME            READY   STATUS    RESTARTS   AGE
configmap-pod   1/1     Running   0          14s
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl exec -it configmap-pod -- printenv
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=configmap-pod
TERM=xterm
config1.txt=key1=value1

config2.txt=key2=value2

config3.txt=key3=value3

KUBERNETES_PORT_443_TCP_PROTO=tcp
KUBERNETES_PORT_443_TCP_PORT=443
KUBERNETES_PORT_443_TCP_ADDR=10.96.0.1
KUBERNETES_SERVICE_HOST=10.96.0.1
KUBERNETES_SERVICE_PORT=443
KUBERNETES_SERVICE_PORT_HTTPS=443
KUBERNETES_PORT=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP=tcp://10.96.0.1:443
NGINX_VERSION=1.29.5
NJS_VERSION=0.9.5
NJS_RELEASE=1~trixie
ACME_VERSION=0.3.1
PKG_RELEASE=1~trixie
DYNPKG_RELEASE=1~trixie
HOME=/root
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8>
```

Step 5: Clean Up

After completing the exercise, clean up the resources created:

kubectl delete pod folder-configmap-pod

kubectl delete configmap folder-configmap

rm -r configmap-folder

For PowerShell: - Remove-Item configmap-folder -Recurse -Force

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl delete configmap folder-configmap
configmap "folder-configmap" deleted from default namespace
```

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl delete pod configmap-pod
pod "configmap-pod" deleted from default namespace
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl get pods
No resources found in default namespace.
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8>
```

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> Remove-Item configmap-folder -Recurse -Force
```

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> Test-Path configmap-folder
False
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8>
```